

OUR PERFORMANCE

We continue to make progress on our energy intensity reduction goal through the use of more efficient processes as well as our investments of waste heat recovery systems and energy-efficient motors and lighting. While our overall energy use increased in 2018, our energy intensity improved by 1.2% from 2017. Much of this

improvement can be attributed to implementation of more efficient processes in our Purification Solutions business. We recognize our progress on this goal has been slow, but we look to current and planned energy center investments as well as energy audits to achieve this goal.

GRI 302-1 | GRI 302-3

ENERGY USE

Exclusive of Energy Sold

USAGE MMGJ		INTENSITY* Intensity GJ / MT of Production	
2018	136.39	2018	62.22
2017	131.16	2017	62.99
2016	126.13	2016	63.10
			

*Progress Toward 2025 Goal **44%**



ENERGY TYPES

0.1%	LIQUID FUELS MMGJ	0.09
0.1%	STEAM MMGJ	0.12
1.8%	PURCHASED ELECTRICITY MMGJ	2.65
2.9%	NATURAL GAS MMGJ	4.18
95.1%	RAW MATERIALS MMGJ	137.32
TOTAL MMGJ		144.36

Optimizing energy usage in Sarnia

As part of our pursuit of efficient technologies, our facility in Sarnia, Canada, successfully implemented several measures to improve energy efficiency. The facility converted approximately 100 light fixtures from high-pressure sodium lights to LED lights, saving an estimated 40,500 kilowatt hours (kWh) of electricity per year. Additionally, over a two-year period, the facility initiated several improvements to reduce its usage of natural gas, including the addition of new building insulation to reduce heat loss and the installation of several new electric steam condensate pumps that allow for the recovery of hot water. The combined savings from the improvement projects was approximately 1,800,000 m³ of avoided natural gas consumption, an amount equivalent to that used by approximately 375 households in Canada over the course of a year.

Converting waste heat to energy in Shimonoseki

Our Shimonoseki, Japan, site recently found a new way to convert excess steam to electricity, reducing our dependency on purchased electricity. The facility is required to incinerate 100% of tail gas produced by the carbon black manufacturing process and convert the heat to steam. With the installation of three STEAMSTAR™ generators, we can now convert that steam to electrical power. Each machine can generate up to 160 kilowatts of energy, and the steam that cannot be used internally is sold to a neighboring facility.



EMISSIONS

OUR COMMITMENT

We understand the impacts associated with GHG emissions and other air pollutants, including linkages to climate change, public health concerns and threats to ecosystems. We also recognize that emission reductions are often a signal of improving overall operational efficiency at our manufacturing facilities. We therefore see our focus on emission reductions as a potential competitive advantage, and we look to partner with our customers and others who share our goal of reducing emissions impacts across the value chain. We monitor our emissions closely and work to minimize our respective impacts in support of maintaining compliance with our regulatory obligations and reducing our global environmental footprint.

GRI 103-1

OUR MANAGEMENT APPROACH

The overall management strategy for our emission reduction program is led by our business segments, regions and individual facilities. We continue to look to develop and implement cost-effective technologies that help meet both our emission reduction and our operational efficiency goals. We engage closely with governments in the areas where we operate, as they aim to reduce industrial and commercial sources of emissions in their communities. Often, this engagement is through our active participation and leadership in various industry groups.

GRI 103-2 | GRI 103-3

OUR PERFORMANCE

By the end of 2018, we had realized 50% of our 2025 GHG reduction target. This was achieved through efforts to capture waste heat and offset our purchase of electricity and by improving our process yields through technology, product selection and feedstock selection. Looking forward, additional process technology improvements are under development for deployment at one of our recently announced expansions in Asia, with the expectation that this technology could be deployed at other facilities.

GHG EMISSIONS	2016	2017	2018
ABSOLUTE EMISSIONS			
SCOPE 1 MM MT CO ₂ e	4.50	4.58	4.77
SCOPE 2 MM MT CO ₂ e	0.34	0.33	0.32
EMISSION INTENSITY*			
SCOPE 1 MT CO ₂ e / MT Production	2.25	2.20	2.18
SCOPE 2 MT CO ₂ e / MT Production	0.17	0.16	0.15

*Progress Toward 2025 Goal **50%**

SO ₂ EMISSIONS	2016	2017	2018
SO₂ ABSOLUTE EMISSIONS KMT	41.80	43.20	43.73
SO₂ EMISSION INTENSITY* MT / KMT Production	20.91	20.75	19.95

*Progress Toward 2025 Goal **30%**

NO _x EMISSIONS	2016	2017	2018
NO_x ABSOLUTE EMISSIONS KMT	13.97	13.23	9.77
NO_x EMISSION INTENSITY* MT / KMT Production	6.99	6.35	4.45

*Goal Achieved

In 2018, we experienced our first full year of operations of our NO_x emission reduction system at our facility in Pampa, Texas, USA. This system has continuously achieved a 90% reduction for what was previously our most intensive facility in terms of NO_x emissions. Furthermore, we successfully completed the NO_x control system at our largest carbon black manufacturing facility, in Tianjin, China; these controls also achieved a 90% reduction of NO_x emissions. Through these efforts, we were able to exceed our global NO_x emission reduction target well ahead of schedule.

Our SO₂ emissions were consistent with previous years, with slight variation based on fluctuations in our feedstock sulfur content and the products we produced. However, we continue to make progress on the implementation of our SO₂ control system at our facility in Franklin, Louisiana, USA, and we have started the early engineering for the control system at our nearby Ville Platte facility. These systems are expected to be operational in 2021 and 2022, respectively.

GRI 305-1 | GRI 305-2 | GRI 305-4 | GRI 305-7



WASTE & SPILLS

OUR COMMITMENT

We acknowledge the potential impact that solid waste disposal or spills of hazardous materials could have on the environment in our local communities. Therefore, we take a targeted approach to minimize waste, and we work toward zero spills at our facilities. We also believe that finding alternative uses for waste materials and byproducts presents opportunities to further contribute to a circular economy. As such, our waste goal is to reduce the amount of waste generated at our facilities that is not beneficially used for energy recovery or other purposes.

GRI 103-1

OUR MANAGEMENT APPROACH

Waste is measured by tracking progress against our intensity target and seeking opportunities to reduce waste during manufacturing. Expectations for the management of waste and the handling of chemicals are governed by internal standards applicable to all our manufacturing facilities. Compliance with these expectations is regularly evaluated through our comprehensive SH&E audit program. We aim to avoid all spills through careful management and design of the processes at our facilities, and we conduct training on spill management and emergency response so we are prepared in the event of an incident. We also implement programs to identify reuse opportunities for off-quality products and other waste streams, and to improve facility resource efficiency to reduce waste generation.

GRI 103-2 | GRI 103-3

OUR PERFORMANCE

Our goal for reducing the intensity of disposed waste was achieved in 2016, but in 2018 we continued to look for ways to reduce waste generation and beneficially use the waste we do generate. Because a one-time sludge removal project in 2017 increased the overall amount of waste generated, we removed the quantity of waste generated from that project from year-over-year trending to provide a better evaluation of progress within our ongoing operations. Taking this into account, we reduced our waste disposal intensity by 2.1% over 2017 and by 2.6% over 2016. In 2018, we also saw 17% and 25% increases in the amount of waste we were able to beneficially use over 2017 and 2016, respectively. The majority of this increase was related to material from tank cleanings which was reused for energy recovery.

A single waste stream at one facility accounts for over 93% of our disposed waste. This stream is directly related to the production activities at that facility, and we expect planned changes will likely reduce that waste stream significantly in the future.

In 2018, we experienced a release at our facility in Franklin, Louisiana, USA, as the result of a relief valve that did not properly close upon unit startup, triggering notifications

to state and federal regulatory agencies. There were no injuries as a result of this event. Corrective actions were implemented to avoid this issue in the future, including upgrading the relief valve and installing additional alarms and interlocks to prevent operation of the unit with this valve open.

GRI 306-2 | GRI 306-3

WASTE GENERATION	2016	2017	2018
ABSOLUTE VOLUME			
NONHAZARDOUS KMT	44.98	102.65	49.83
HAZARDOUS KMT	378.05	398.28	405.86
WASTE INTENSITY			
NONHAZARDOUS MT / KMT Production	22.50	49.30	22.73
HAZARDOUS MT / KMT Production	189.11	191.27	185.15



WASTE DISPOSAL	2016	2017	2018
TOTAL WASTE DISPOSED – ABSOLUTE KMT	401.34	415.78	428.62
TOTAL WASTE DISPOSED INTENSITY* MT / KMT Production	200.76	199.68	195.53
*Goal Achieved			
BENEFICIALLY REUSED MATERIALS	2016	2017	2018
WASTE BENEFICIALLY USED KMT	21.69	85.16	27.07
WASTE BENEFICIALLY USED INTENSITY MT / KMT Production	10.85	40.90	12.35

Meeting recycling goals in Haverhill

As part of a continued effort to reduce waste where possible, our facility in Haverhill, Massachusetts, USA, set a target for the 2018 fiscal year of recycling 50 tons of materials and 5 million gallons of water. The facility team implemented water recovery through reverse osmosis and was able to recycle 5.6 million gallons of water. By employing proper waste segregation and working with key partners, the facility was able to recycle 54 tons of cardboard, paper, plastics, metal, wood pallets and drums. Both of these efforts exceeded the facility's 2018 targets and set up the team for continuous improvement in future years.

MATERIAL TOPICS – ACTING RESPONSIBLY FOR THE PLANET

Environmental Compliance

GRI 103: Management Approach 2016

Disclosure Number	Disclosure Title	Page Number / Response
103-1	Explanation of the material topic and its boundaries	pp. 8, 10
103-2	The management approach and its components	pp. 19-20
103-3	Evaluation of the management approach	pp. 19-20

GRI 307: Environmental Compliance 2016

Disclosure Number	Disclosure Title	Page Number / Response
307-1	Non-compliance with environmental laws and regulations	p. 20 Adhering to local environmental laws and regulations is the responsibility of facility general managers as well as site environmental managers located at each facility. In support of compliance efforts, resources include a robust database to track near-miss and ENC events and corrective actions, as well as over \$44.6MM in capital spending in FY 2018 that was dedicated to improving facilities and reducing ENCs.

Energy

GRI 103: Management Approach 2016

Disclosure Number	Disclosure Title	Page Number / Response
103-1	Explanation of the material topic and its boundaries	pp. 8, 19-20
103-2	The management approach and its components	p. 20
103-3	Evaluation of the management approach	p. 20

GRI 302: Energy 2016

Disclosure Number	Disclosure Title	Page Number / Response
302-1	Energy consumption within the organization	p. 21 For more information about our energy and fuel sources, refer to our response to the CDP Climate Change Questionnaire (cdp.net).
302-3	Energy intensity	p. 21 Energy consumption includes all forms of energy consumed by facilities under Cabot's operational control and excludes energy sold to third parties.

MATERIAL TOPICS — ACTING RESPONSIBLY FOR THE PLANET *continued*

Air Pollutants / GHG

GRI 103: Management Approach 2016

Disclosure Number	Disclosure Title	Page Number / Response
103-1	Explanation of the material topic and its boundaries	pp. 8, 19, 22
103-2	The management approach and its components	p. 22
103-3	Evaluation of the management approach	p. 22

GRI 305: Emissions 2016

Disclosure Number	Disclosure Title	Page Number / Response
305-1	Direct (Scope 1) GHG emissions	pp. 22-23 Our greenhouse gas calculations were completed in accordance with <i>The Greenhouse Gas Protocol: A Corporate Accounting and Reporting Standards</i> (Revised Edition), drawing guidance from the <i>IPCC Guidelines for National Greenhouse Gas Inventories – 2006</i> and <i>The Climate Registry: General Reporting Protocol</i> . Emissions were calculated using the operational control approach and IPCC Second Assessment Report 100-year global warming potentials, and included emissions of CO ₂ , CH ₄ and N ₂ O. We maintain databases that track monthly usage volumes of feedstock materials and fossil fuels, as well as production volume. Our 2017 and 2018 GHG emissions were verified in alignment with the principles of the <i>International Standard on Assurance Engagements ISAE 3000</i> (Revised) under a Limited Level of Assurance by ERM CVS.
305-2	Indirect (Scope 2) GHG emissions	pp. 22-23 See Disclosure 305-1 in the GRI Content Index for a description of GHG monitoring methods.
305-4	GHG emissions intensity	pp. 22-23 GHG intensity is calculated as MT CO ₂ e emissions/MT of product. The intensity of our GHG emissions is calculated for all Scope 1 and 2 emissions produced by facilities under Cabot's operational control, as reported under Disclosures 305-1 and 305-2.
305-7	Nitrogen oxides (NO _x), sulfur oxides (SO _x) and other significant air emissions	pp. 22-23 Data reported has been calculated using actual test measurements based on country-specific or U.S. EPA methods, Cabot engineering estimates, or similar emission factors.

Waste & Spills

GRI 103: Management Approach 2016

Disclosure Number	Disclosure Title	Page Number / Response
103-1	Explanation of the material topic and its boundaries	pp. 8, 19, 23
103-2	The management approach and its components	p. 23
103-3	Evaluation of the management approach	p. 23

MATERIAL TOPICS — ACTING RESPONSIBLY FOR THE PLANET *continued*

Waste & Spills

GRI 306: Effluents and Waste 2016

Disclosure Number	Disclosure Title	Page Number / Response																					
306-2	Waste by type and disposal method	pp. 23-24 More than a third of nonhazardous waste generated is recycled or reused, and the majority of hazardous waste is generated at one location and disposed of via on-site deep well injection.																					
<table> <tr> <th>2018 Disposal Methods</th><th>Nonhazardous Waste</th><th>Hazardous Waste</th></tr> <tr> <td>Reused or recycled</td><td>36.7%</td><td>0.6%</td></tr> <tr> <td>Incinerated with energy recovery</td><td>7.4%</td><td>0.6%</td></tr> <tr> <td>Incinerated without energy recovery</td><td>0.4%</td><td>0.2%</td></tr> <tr> <td>Deep well injection</td><td>0.0%</td><td>98.4%</td></tr> <tr> <td>Landfilled</td><td>54.5%</td><td>0.2%</td></tr> <tr> <td>Other</td><td>1.0%</td><td>0.0%</td></tr> </table>			2018 Disposal Methods	Nonhazardous Waste	Hazardous Waste	Reused or recycled	36.7%	0.6%	Incinerated with energy recovery	7.4%	0.6%	Incinerated without energy recovery	0.4%	0.2%	Deep well injection	0.0%	98.4%	Landfilled	54.5%	0.2%	Other	1.0%	0.0%
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306-3	Total number and volume of significant spills	pp. 23-24																					

Water

GRI 103: Management Approach 2016

Disclosure Number	Disclosure Title	Page Number / Response
103-1	Explanation of the material topic and its boundaries	pp. 8, 19, 25
103-2	The management approach and its components	p. 25
103-3	Evaluation of the management approach	p. 25

GRI 303: Water 2016

Disclosure Number	Disclosure Title	Page Number / Response
303-1	Water withdrawal by source	p. 25
303-3	Water recycled and reused	p. 25

GRI 306: Effluents and Waste 2016

Disclosure Number	Disclosure Title	Page Number / Response
306-1	Water discharge by quality and destination	p. 25